NAME: _____ Geometry (FL B.E.S.T.)

6.4 Solving Right Triangles

"Solving" a right triangle means finding every missing side and acute angle. Use the **Pythagorean Theorem** when two sides are known and you need the third. Use a **trig ratio** (SOH-CAH-TOA) when one side and one acute angle are known and you need another side. Use an **inverse trig function** ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) when two sides are known and you need an angle — for example, if $\tan\theta = \frac{3}{4}$, then $\theta = \tan^{-1}(\frac{3}{4})$. Once one acute angle is known, the other follows immediately since the two acute angles are complementary (90° minus the first). This worksheet combines all three tools, and several problems ask you to fully solve a triangle — finding all three sides and both acute angles.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Legs 5 and 12. Find c.

2. One acute angle is 41°. Find the other.

3. Hypotenuse $= 25$, angle $= 30^\circ$. Find the opposite side (exact).

4. Opposite $= 7$, adjacent $= 24$. Express θ.

5. Hypotenuse $= 14$, angle $= 60^\circ$. Find the adjacent side (exact).

6. Adjacent $= 8$, angle $= 45^\circ$. Find the hypotenuse (exact).

7. Opposite $= 9$, hypotenuse $= 15$. Express θ.

8. Legs 20 and 21. Find the hypotenuse.

9. Adjacent $= 12$, $\tan\theta = \frac{5}{12}$. Find the opposite side.

10. Angle $A = 67^\circ$ in a right triangle. Find angle B.

11. Adjacent $= 8$, hypotenuse $= 17$. Express θ.

12. Hypotenuse $= 30$, angle $= 45^\circ$. Find the opposite side (exact).

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13. A right triangle has legs 10 and 24. Find the hypotenuse and both acute angles (as inverse-sine expressions).

14. A right triangle has hypotenuse 34 and one acute angle $28°$. Find both legs (in terms of $\sin 28°$ and $\cos 28°$), then the other acute angle.

15. A right triangle is similar to a 3-4-5 triangle, scaled so its hypotenuse is 45. Find both legs, and express one acute angle as an inverse-sine ratio in lowest terms.

✓ Answer Key

1. 13
2. $49°$
3. 12.5
4. $\tan^{-1}\left(\frac{7}{24}\right)$
5. 7
6. $8\sqrt{2}$
7. $\sin^{-1}\left(\frac{3}{5}\right)$
8. 29
9. 5
10. $23°$
11. $\cos^{-1}\left(\frac{8}{17}\right)$
12. $15\sqrt{2}$
13. Hypotenuse $= 26$; angles $= \sin^{-1}\left(\frac{5}{13}\right)$ and $\sin^{-1}\left(\frac{12}{13}\right)$
14. Opposite leg $= 34\sin 28°$; adjacent leg $= 34\cos 28°$; other angle $= 62°$
15. Legs $= 27$ and 36; one acute angle $= \sin^{-1}\left(\frac{3}{5}\right)$